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```
import pandas as pd
from sklearn.metrics import accuracy_score
from sklearn.tree import DecisionTreeClassifier
from google.colab import drive
drive.mount("/content/drive")
```

```
Mounted at /content/drive
```

```
path="/content/drive/MyDrive/ML/daily_weather.csv"
df=pd.read_csv(path)
df.head(5)
```

	number	air_pressure_9am	air_temp_9am	avg_wind_direction_9am	avg_win
0	0	918.060000	74.822000	271.100000	
1	1	917.347688	71.403843	101.935179	
2	2	923.040000	60.638000	51.000000	
3	3	920.502751	70.138895	198.832133	
4	4	921.160000	44.294000	277.800000	

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1095 entries, 0 to 1094
Data columns (total 11 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   number                  1095 non-null   int64
 1   air_pressure_9am        1092 non-null   float64
 2   air_temp_9am            1090 non-null   float64
 3   avg_wind_direction_9am  1091 non-null   float64
 4   avg_wind_speed_9am      1092 non-null   float64
 5   max_wind_direction_9am  1092 non-null   float64
 6   max_wind_speed_9am      1091 non-null   float64
 7   rain_accumulation_9am   1089 non-null   float64
 8   rain_duration_9am       1092 non-null   float64
 9   relative_humidity_9am   1095 non-null   float64
 10  relative_humidity_3pm   1095 non-null   float64
dtypes: float64(10), int64(1)
memory usage: 94.2 KB
```

```
df.shape
```

```
(1095, 11)
```

```
df.describe()
```

	number	air_pressure_9am	air_temp_9am	avg_wind_direction_9am
count	1095.000000	1092.000000	1090.000000	1091.000000
mean	547.000000	918.882551	64.933001	142.235511
std	316.243577	3.184161	11.175514	69.137859
min	0.000000	907.990000	36.752000	15.500000
25%	273.500000	916.550000	57.281000	65.972506
50%	547.000000	918.921045	65.715479	166.000000
75%	820.500000	921.160073	73.450974	191.000000
max	1094.000000	929.320000	98.906000	343.400000

```
df.columns
```

```
Index(['number', 'air_pressure_9am', 'air_temp_9am',
'avg_wind_direction_9am',
       'avg_wind_speed_9am', 'max_wind_direction_9am',
'max_wind_speed_9am',
       'rain_accumulation_9am', 'rain_duration_9am',
'relative_humidity_9am',
       'relative_humidity_3pm'],
      dtype='object')
```

```
df.isnull().any().any()
```

```
np.True_
```

```
df.isnull().sum()
```

	0
number	0
air_pressure_9am	3
air_temp_9am	5
avg_wind_direction_9am	4
avg_wind_speed_9am	3
max_wind_direction_9am	3
max_wind_speed_9am	4
rain_accumulation_9am	6
rain_duration_9am	3
relative_humidity_9am	0
relative_humidity_3pm	0

dtype: int64

```
#print missing value
df[df.isnull().any(axis=1)]
```

	number	air_pressure_9am	air_temp_9am	avg_wind_direction_9am	avg
16	16	917.890000	NaN	169.200000	
111	111	915.290000	58.820000	182.600000	
177	177	915.900000	NaN	183.300000	
262	262	923.596607	58.380598	47.737753	
277	277	920.480000	62.600000	194.400000	
334	334	916.230000	75.740000	149.100000	
358	358	917.440000	58.514000	55.100000	
361	361	920.444946	65.801845	49.823346	
381	381	918.480000	66.542000	90.900000	
409	409	NaN	67.853833	65.880616	
517	517	920.570000	53.600000	100.100000	
519	519	916.250000	55.670000	176.400000	
546	546	NaN	42.746000	251.100000	
620	620	921.200000	56.786000	192.300000	
625	625	912.400000	50.774000	171.600000	
656	656	920.830000	66.344000	NaN	
670	670	910.920000	48.362000	156.500000	
672	672	922.448945	72.863773	NaN	
705	705	911.900000	59.072000	199.800000	
731	731	922.970167	51.391847	33.810942	
737	737	917.895130	76.804690	104.771020	
788	788	917.923442	73.249717	42.101739	
840	840	918.043767	NaN	181.774042	
848	848	915.250000	37.562000	246.500000	
861	861	919.065408	NaN	172.303728	
869	869	NaN	45.104000	259.000000	
998	998	914.140000	71.240000	NaN	
1031	1031	922.669195	NaN	47.946284	
1035	1035	919.670000	77.576000	171.800000	
1063	1063	917.300185	65.790001	NaN	
1066	1066	919.564869	73.726732	68.704694	

```
df.drop('number', axis=1, inplace=True, errors='ignore')
df.columns
```

```
Index(['air_pressure_9am', 'air_temp_9am', 'avg_wind_direction_9am',
       'avg_wind_speed_9am', 'max_wind_direction_9am',
'max_wind_speed_9am',
       'rain_accumulation_9am', 'rain_duration_9am',
'relative_humidity_9am',
       'relative_humidity_3pm'],
      dtype='object')
```

```
df.shape
```

```
(1095, 10)
```

```
before_rows=df.shape[0]
df = df.dropna()
after_rows = df.shape[0]
print("Before rows:",before_rows)
print("After rows:",after_rows)
print("The number of dropped rows are:",before_rows-after_rows)
```

```
Before rows: 1095
After rows: 1064
The number of dropped rows are: 31
```

```
df.shape
```

```
(1064, 10)
```

```
clean_data = df.copy()
clean_data ['high_humidity_label'] = (clean_data['relative_humidity_3pm']
print(clean_data['high_humidity_label'])
```

```
0       1
1       0
2       0
3       0
4       1
       ..
1090    1
1091    1
1092    1
1093    1
1094    0
Name: high_humidity_label, Length: 1064, dtype: int64
```

```
clean_data.columns
```

```
Index(['air_pressure_9am', 'air_temp_9am', 'avg_wind_direction_9am',
       'avg_wind_speed_9am', 'max_wind_direction_9am',
'max_wind_speed_9am',
       'rain_accumulation_9am', 'rain_duration_9am',
'relative_humidity_9am',
```

```
       'relative_humidity_3pm', 'high_humidity_label'],
      dtype='object')
```

```
clean_data.drop('relative_humidity_9am', axis=1, inplace=True, errors='ig
clean_data.drop('relative_humidity_3pm', axis=1, inplace=True, errors='ig
clean_data.columns
```

```
Index(['air_pressure_9am', 'air_temp_9am', 'avg_wind_direction_9am',
       'avg_wind_speed_9am', 'max_wind_direction_9am',
'max_wind_speed_9am',
       'rain_accumulation_9am', 'rain_duration_9am',
'high_humidity_label'],
      dtype='object')
```

```
X=clean_data.iloc[:,:-1]
print(X)
```

```
      air_pressure_9am  air_temp_9am  avg_wind_direction_9am  \
0           918.060000     74.822000              271.100000
1           917.347688     71.403843              101.935179
2           923.040000     60.638000               51.000000
3           920.502751     70.138895              198.832133
4           921.160000     44.294000              277.800000
...                ...           ...                     ...
1090        918.900000     63.104000              192.900000
1091        918.710000     49.568000              241.600000
1092        916.600000     71.096000              189.300000
1093        912.600000     58.406000              172.700000
1094        921.530000     77.702000               97.100000

      avg_wind_speed_9am  max_wind_direction_9am  max_wind_speed_9am  \
0               2.080354              295.400000            2.863283
1               2.443009              140.471549            3.533324
2              17.067852               63.700000           22.100967
3               4.337363              211.203341            5.190045
4               1.856660              136.500000            2.863283
...                  ...                     ...                 ...
1090            3.869906              207.300000            5.212070
1091            1.811921              227.400000            2.371156
1092            3.064608              200.800000            3.892276
1093            3.825167              189.100000            4.764682
1094            3.265932              125.900000            4.451511

      rain_accumulation_9am  rain_duration_9am
0                       0.0                0.0
1                       0.0                0.0
2                       0.0               20.0
3                       0.0                0.0
4                       8.9            14730.0
...                     ...                ...
1090                    0.0                0.0
1091                    0.0                0.0
1092                    0.0                0.0
1093                    0.0                0.0
1094                    0.0                0.0

[1064 rows x 8 columns]
```

```
Y=clean_data.iloc[:,-1]
print(Y)
```

```
0       1
1       0
2       0
3       0
4       1
       ..
1090    1
1091    1
1092    1
1093    1
1094    0
Name: high_humidity_label, Length: 1064, dtype: int64
```

```
from sklearn.model_selection import train_test_split
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.3,
```

```
X_train
```

	air_pressure_9am	air_temp_9am	avg_wind_direction_9am	avg_wind_spe
5	915.30000	78.404000	182.800000	9.
315	917.09000	63.266000	175.500000	6.
918	915.07000	58.046000	186.200000	7.
105	917.40000	73.868000	181.700000	4.
98	916.52000	51.638000	190.800000	7.
...	...	...	...	
336	912.89000	45.320000	213.400000	21.
476	917.18000	62.834000	186.300000	3.
123	920.18000	62.744000	32.600000	3.
1075	922.85811	64.989361	63.483047	10.
886	918.70000	69.422000	161.100000	0.

744 rows × 8 columns

```
X_test
```

	air_pressure_9am	air_temp_9am	avg_wind_direction_9am	avg_wind_spe
32	918.700000	76.226000	184.600000	4.
1084	917.130000	80.240000	183.300000	1.
423	921.090000	70.124000	182.800000	2.
767	915.220000	73.292000	148.300000	1.
818	921.400000	63.050000	223.200000	1.
...	...	...	...	
503	919.738161	67.432744	147.470881	1.
585	916.250000	79.754000	48.600000	1.
389	916.500000	74.588000	101.200000	1.
715	923.600000	65.030000	62.300000	7.
352	925.280000	66.470000	49.400000	11.

320 rows × 8 columns

```
Y_train
```

	high_humidity_label
5	1
315	1
918	1
105	1
98	1
...	...
336	1
476	1
123	1
1075	0
886	0

744 rows × 1 columns

dtype: int64

```
Y_test
```

	high_humidity_label
32	1
1084	1
423	1
767	1
818	1
...	...
503	0
585	1
389	1
715	1
352	0

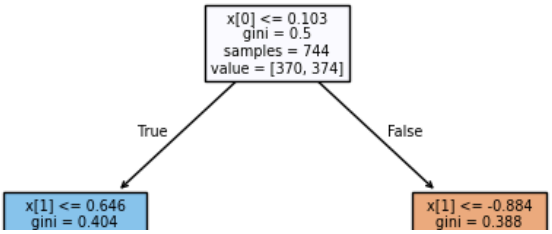
320 rows × 1 columns

dtype: int64

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler = scaler.transform(X_test)
```

```
humidity_classifier = DecisionTreeClassifier(max_leaf_nodes=10, random_st
humidity_classifier.fit(X_train, Y_train)
```

```
▾ DecisionTreeClassifier
DecisionTreeClassifier(max_leaf_nodes=10, random_state=0)
```

```
import matplotlib.pyplot as plt
from sklearn import tree
plt.figure(figsize=(10,10))
tree.plot_tree(humidity_classifier, filled=True)
```

```
0.201\nsamples = 230\nvalue = [26, 204]'),
 Text(0.07142857142857142, 0.4166666666666667, 'gini = 0.499\nsamples =
25\nvalue = [12, 13]'),
 Text(0.21428571428571427, 0.4166666666666667, 'gini = 0.127\nsamples =
205\nvalue = [14.0, 191.0]'),
 Text(0.2857142857142857, 0.5833333333333334, 'gini = 0.499\nsamples =
161\nvalue = [84, 77]'),
 Text(0.5714285714285714, 0.75, 'x[1] <= -0.884\ngini = 0.388\nsamples =
353\nvalue = [260, 93]'),
 Text(0.4821428571428571, 0.8333333333333333, '  False'),
 Text(0.42857142857142855, 0.5833333333333334, 'x[2] <= -1.287\ngini =
0.451\nsamples = 64\nvalue = [22, 42]'),
 Text(0.35714285714285715, 0.4166666666666667, 'x[3] <= -0.155\ngini =
0.298\nsamples = 22\nvalue = [18, 4]'),
 Text(0.2857142857142857, 0.25, 'gini = 0.0\nsamples = 4\nvalue = [0,
4]'),
 Text(0.42857142857142855, 0.25, 'gini = 0.0\nsamples = 18\nvalue = [18,
0]'),
 Text(0.5, 0.4166666666666667, 'gini = 0.172\nsamples = 42\nvalue = [4,
38]'),
 Text(0.7142857142857143, 0.5833333333333334, 'x[2] <= -0.754\ngini =
0.291\nsamples = 289\nvalue = [238, 51]'),
 Text(0.6428571428571429, 0.4166666666666667, 'gini = 0.114\nsamples =
149\nvalue = [140, 9]'),
 Text(0.7857142857142857, 0.4166666666666667, 'x[1] <= -0.552\ngini =
0.42\nsamples = 140\nvalue = [98, 42]'),
 Text(0.7142857142857143, 0.25, 'gini = 0.0\nsamples = 7\nvalue = [0,
7]'),
 Text(0.8571428571428571, 0.25, 'x[0] <= 0.504\ngini = 0.388\nsamples =
133\nvalue = [98, 35]'),
 Text(0.7857142857142857, 0.08333333333333333, 'gini = 0.497\nsamples =
54\nvalue = [29.0, 25.0]'),
 Text(0.9285714285714286, 0.08333333333333333, 'gini = 0.221\nsamples =
79\nvalue = [69, 10]')]
```